

Hazard Analysis Flexin

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Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
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1 Introduction

This document aims to illustrate the hazards that may occur during the development and deployment of Flexin'. It defines hazards for this project, the purpose of analyzing them, and how the team will mitigate them to ensure safe use of Flexin'.

1.1 Flexin' Problem Statement

Personal finance and fitness follow the same pattern: the payoff for effort put in today shows up years later. Someone who invests \$100 a month from age 25, assuming a 7% annual return, contributes \$48,000 over 40 years and ends up with roughly \$262,000. Waiting until 35 to start the same plan leaves them with about \$122,000. The same logic applies to exercise. A single walk or workout changes very little, but the habit, kept up for years, is what separates a healthy 40-year-old from an unhealthy one. In both cases the early actions matter most, and in both cases they are the easiest to skip, because the reward is too far away to feel real.

The concepts behind this are not widely understood, and young people understand them the least. A 2012 survey of 6,911 Canadians asked three basic questions covering compound interest, inflation, and risk diversification [1]. Only 42% of respondents answered all three correctly. Among respondents under 35, that number fell to about 30%, the lowest of any age group, and 44% of them answered “don't know” to at least one question. Only about half of respondents under 35 knew that buying a single company's stock is usually riskier than buying a stock mutual fund. The same study found that respondents who answered all three questions correctly were 10 percentage points more likely to have retirement savings, which suggests that this gap in knowledge carries through to real financial outcomes.

Tools for practising each skill already exist, but separately. Stock market simulators let people trade with fake money, and fitness apps reward steps with points and badges. A market simulator only helps someone who is already motivated to learn about investing, which excludes most of the young adults the survey shows are furthest behind. Fitness rewards, meanwhile, rarely lead to anything beyond the app itself. No product currently connects the two so that a habit a user is already building (or wants to build) becomes their way into learning how money grows.

1.2 Hazard Analysis Introduction

A hazard as it pertains to Flexin' is a condition in the software that can create an unwanted effect of event on Flexin's users, or the general environment. In software, these hazards stem from poor design decisions or requirements. These hazards must be addressed prior to and during development in order to limit any unwanted impact on users.

As described above, Flexin' focuses on developing healthy habits for investing and fitness and teach users the effect of compounding. However, to accomplish this goal there are many hazards that may hurt users physically, psychologically, or financially. These hazards may come from poor software implementation, incompatible data, or faulty integration with outside sources. The hazard analysis document below aims to identify all hazards and how the team plans to address them.

2 Scope and Purpose of Hazard Analysis

The hazard analysis document outlines the hazards that may occur while using Flexin'. The high level hazards in scope of Flexin' include:

- **Physical/Psychological Harm:** Includes injuries that may occur during workouts that users intend to track. Also includes any impacts on mental health caused by the platform, such as addiction (similar to Instagram or TikTok) or feelings of embarrassment/shame for missing goals.
- **Financial Loss:** It is imperative to note that any interactions with financial markets on Flexin' are not indicative of real market results. Users may take strategies learned on the platform and incur serious financial loss.
- **Data Mismanagement:** Poor management of user data from database or cybersecurity perspectives may result in data leaks or loss of data. In addition, users improperly providing data can cause over/under rewarding of virtual currency, and can therefore lead to poor learning.

The purpose of the hazard analysis is to identify and mitigate risks that could affect the reliability, correctness and safety of the Flexin' app. Conduction of this analysis will allow the team to anticipate failures early, and design around the hazards in a way that prioritizes the safety of the users. The findings from this process will inform safety and security requirements, as well as having an impact on compliance and maintainability requirements. In addition, this work will contribute to the creation of a fun, safe, and intuitive platform for users to learn about compounding and the importance of fitness and financial literacy.

3 System Boundaries and Components

Flexin' will consist of many key components, all of which will come together to create the product. These components may not be separate entities, but they are distinct enough to present their own hazards.

- **Component 1: Account Creation & Authentication**
 - **Purpose:** Manages account creation and authentication for users

- **Key Functions:**
 - * Allow users to create accounts for Flexin’
 - * Populate the database with the user account to store the created account
 - * Store authentication key to ensure that only the account owner can log into their Flexin’ account
- **Component 2: User Information**
 - **Purpose:** Manages user profile data such as name, height, weight, goals, workout data, etc.
 - **Key Functions:**
 - * Read and write to the database with user information
 - * Provide profile data for users
 - * Allow users to set preferences within Flexin’
- **Component 3: Fitness Device Integrations**
 - **Purpose:** Allow Flexin’ to connect with users’ fitness devices
 - **Key Functions:**
 - * Connect with users’ devices
 - * Take data from said devices when relevant
 - * Ensure that connections with the devices are secure and reliable
- **Component 4: Exercise Data**
 - **Purpose:** Track and analyze users’ exercise data
 - **Key Functions:**
 - * Interact with the data collected from the fitness devices
 - * Analyze exercise data against user goals
 - * Read and write to the database with results and analytics
- **Component 5: Currency Attribution**
 - **Purpose:** Turn exercise results into virtual currency
 - **Key Functions:**
 - * Use the analytics from exercise data
 - * Compute the correct amount of currency to reward users
 - * Read and write to the database with users’ balances
- **Component 6: Market Simulation**
 - **Purpose:** The underlying behaviour of the market in Flexin’
 - **Key Functions:**

- * Teach users about the effects of compounding
- * Mimic proper stock market behaviour
- * Incorporate randomness through seeding
- **Component 7: Investment Platform**
 - **Purpose:** Allow users to invest in the simulated market with their currency
 - **Key Functions:**
 - * Provide users with securities to invest in
 - * Allow for buying and selling of said securities
 - * Read and write to the database with investment information
- **Component 8: Social Management**
 - **Purpose:** Create a space for users to compete with their friends on Flexin'
 - **Key Functions:**
 - * Manage group creation, validation and invites
 - * Use the simulated markets randomness to create fair competition
 - * Read and write to the database with group information
- **Component 9: Database**
 - **Purpose:** The single source of truth for all aspects of Flexin'
 - **Key Functions:**
 - * ACID database that will ensure all data is correct
 - * Accept reads and writes from components
- **Component 10: User Interface & Experience**
 - **Purpose:** Provide users a place to interact with all backend components of Flexin'
 - **Key Functions:**
 - * Manage all user facing screens, navigation, and input
 - * Ensure that components connect well to create a simple experience

4 Critical Assumptions

[These assumptions that are made about the software or system. You should minimize the number of assumptions that remove potential hazards. For instance, you could assume a part will never fail, but it is generally better to include this potential failure mode. —SS]

5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

8 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

8.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

8.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

8.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

9 Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

References

- [1] D. Boisclair, A. Lusardi, and P.-C. Michaud, "Financial literacy and retirement planning in Canada," *Journal of Pension Economics and Finance*, vol. 16, no. 3, pp. 277–296, 2017, doi: 10.1017/S1474747215000311.